

Software Requirements Document (SRD)

Project Title

Industrial Asset & Installation Management Platform

Client

Tayyab Traders

Project Type

Mobile Application (Primary) + Backend API

Built using **React Native** (Frontend) + **Node.js** + **PostgreSQL** (Backend)

1. Project Overview

The purpose of this application is to digitize and centralize:

1. Machine inventory management
2. Shipment tracking
3. Warehouse tracking
4. Installation geo-mapping
5. Video training & documentation management

The application will be used internally by company staff to manage machine lifecycle from warehouse to installation.

This is **NOT an AI-based system**. All data tracking is manual and controlled by internal users.

2. User Roles & Permissions

The system must support **Role-Based Access Control (RBAC)**.

2.1 User Roles

1 Super Admin

- Full system control
- Manage users
- Add/edit/delete machines
- Access all warehouses
- View installation map
- View all shipments
- Upload training videos

2 Warehouse Manager

- Manage machines within assigned warehouse
- Update machine status
- Create shipment entries
- Cannot manage users

3 Sales/Operations Staff

- View inventory
- View shipment status
- View installation locations
- Cannot modify warehouses

4 Technician (Optional)

- View assigned installations
 - Update installation status
 - Add installation images
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3. System Architecture

3.1 Frontend

- React Native (iOS & Android)
- React Navigation
- State Management: Redux or Zustand
- Google Maps integration
- REST API integration

3.2 Backend

- Node.js (Express or NestJS)
- PostgreSQL database
- Prisma or Sequelize ORM
- JWT Authentication
- RESTful API

3.3 Cloud Infrastructure

- AWS / DigitalOcean
 - S3 (for video & document storage)
 - Secure API hosting
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4. MODULE 1 — Inventory & Asset Management

4.1 Purpose

To manage all machines owned by the company and track their lifecycle.

4.2 Machine Data Model

Each Machine must have:

- Unique ID (Auto-generated)
 - Serial Number (Unique)
 - Machine Model
 - Machine Category
 - Purchase Date
 - Cost (Optional)
 - Current Status
 - Current Warehouse
 - Assigned Client (if installed)
 - Installation Location (if installed)
 - Created At
 - Updated At
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4.3 Machine Status Lifecycle

Machine statuses must include:

1. In Warehouse
2. Reserved
3. Under Shipment
4. Delivered
5. Installed
6. Under Maintenance
7. Returned

Status changes must:

- Be logged in history
 - Include date & user who changed it
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4.4 Features

- Add new machine
 - Edit machine details
 - Delete machine
 - View machine list
 - Filter by:
 - Status
 - Warehouse
 - Model
 - Date
 - Search by serial number
 - View full status history
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4.5 History Log

Every machine must have:

- Status change history
- Timestamp
- User who updated
- Optional comment

This ensures traceability.

5. MODULE 2 — Warehouse Management

5.1 Warehouse Data Model

Each warehouse must include:

- Warehouse ID
 - Name
 - Address
 - City
 - Manager
 - Contact Info
 - Capacity (Optional)
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5.2 Features

- Add warehouse
 - Edit warehouse
 - Delete warehouse
 - View machines stored in warehouse
 - View warehouse inventory count
 - Filter by machine type
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6. MODULE 3 — Shipment Tracking

6.1 Purpose

To track movement of machines between warehouses or to client locations.

6.2 Shipment Data Model

Each shipment must include:

- Shipment ID
- Machine ID

- From Warehouse
 - To Warehouse OR Client
 - Shipment Date
 - Expected Delivery Date
 - Delivery Status
 - Delivery Confirmation
 - Updated By
 - Notes
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6.3 Shipment Status

Shipment statuses:

1. Created
2. Dispatched
3. In Transit
4. Delivered
5. Cancelled

When shipment is created:

- Machine status must change to “Under Shipment”

When delivered:

- Machine status must update automatically
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6.4 Features

- Create shipment
 - Edit shipment
 - Mark as delivered
 - View shipment list
 - Filter by date/status
 - Shipment history tracking
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7. MODULE 4 — Installation & Geo-Mapping

7.1 Purpose

To visualize where machines are installed.

7.2 Installation Data Model

Each installation must include:

- Installation ID
 - Machine ID
 - Client Name
 - Client Contact
 - Address
 - City
 - Country
 - Latitude
 - Longitude
 - Installation Date
 - Installed By
 - Images (Optional)
 - Notes
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7.3 Map Features

- Google Maps integration
 - Display pin for each installation
 - Tap pin → show:
 - Machine serial
 - Model
 - Client name
 - Installation date
 - Filter map by:
 - Machine type
 - City
 - Installation date
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7.4 Advanced (Optional Future)

- Cluster markers
- Heatmap of installations

- Territory-based analytics
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8. **MODULE 5 — Video Training & Documentation Hub**

8.1 Purpose

To store training materials for each machine model.

8.2 Data Model

Each Training Entry must include:

- Video ID
 - Title
 - Machine Model
 - Description
 - Video URL (stored in S3 or Vimeo)
 - PDF Manual URL
 - Uploaded By
 - Upload Date
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8.3 Features

- Upload training video
 - Upload PDF manual
 - Categorize by machine model
 - Search training materials
 - Filter by category
 - View video inside app
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9. **Authentication & Security**

9.1 Authentication

- Email + Password login
- JWT token authentication
- Password encryption (bcrypt)

9.2 Authorization

- Role-based permissions
- Middleware validation for protected routes

9.3 Security Requirements

- HTTPS only
 - Secure file storage
 - Prevent unauthorized data access
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10. Dashboard & Reporting

Admin Dashboard Must Show:

- Total machines
- Machines by status
- Machines per warehouse
- Installations count
- Shipments in transit
- Recent activity log

Optional:

- Download CSV reports
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11. Mobile App Requirements

Screens Required

1. Login
2. Dashboard

3. Machine List
 4. Machine Details
 5. Add/Edit Machine
 6. Warehouse List
 7. Shipment List
 8. Installation Map
 9. Training Library
 10. User Management (Admin)
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12. Performance Requirements

- Must support minimum 5,000+ machines
 - Map must load efficiently
 - API response time < 2 seconds
 - Scalable backend design
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13. Non-Functional Requirements

- Clean UI
 - Offline-safe form submissions (optional)
 - Error logging
 - Activity tracking
 - Modular codebase
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14. Project Phases (Recommended)

Phase 1 – Core Asset Management

- Authentication
- Machine CRUD
- Warehouse management
- Status lifecycle
- Basic dashboard

Phase 2 – Shipment + Map

- Shipment tracking
- Installation management
- Google Maps integration

Phase 3 – Training Module

- Video uploads
 - Documentation management
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15. Estimated Timeline (Rough)

If handled by 1–2 developers:

- Phase 1: 4–6 weeks
- Phase 2: 3–4 weeks
- Phase 3: 2–3 weeks

Total: 10–13 weeks

16. Final System Definition

This system should be developed as a:

Mobile-based Industrial Asset Lifecycle & Installation Management Platform

It combines:

- Asset tracking
 - Logistics tracking
 - Geo-visualization
 - Digital training management
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